Faisal Mahmud Abir

AI Software Engineer
faisalabir4121@gmail.com | +880 1787-474121 | LinkedIn | GitHub | Portfolio

PROFILE SUMMARY

Computer Science and Engineering graduate specializing in AI Software Engineering, Machine Learning, and Computer Vision. Experienced in developing AI-powered applications, deep learning models, LLM-based solutions, and scalable backend systems. Passionate about building intelligent software, full-stack applications, and AI-driven automation solutions that solve real-world problems.

TECHNICAL SKILLS

Programming & Data: Python, C++, JavaScript, TypeScript, SQL, NumPy, Pandas, Advanced Excel, Power BI

AI/ML & Computer Vision: PyTorch, Scikit-Learn, Hugging Face Transformers, YOLO, OpenCV, Model Evaluation, XGBoost, Feature Engineering, Model Deployment & Optimization, MLflow

Generative AI & Agents: RAG, LLM Integration, AI Agents, Tool Calling

Full-Stack & Backend: React.js, Next.js, Tailwind CSS, FastAPI, Flask, REST APIs, WebSockets

Databases & Cloud: PostgreSQL, MongoDB, ChromaDB, Neon, AWS, GCP, Docker, Vercel, Git, GitHub, Linux, n8n

PROJECT EXPERIENCE

Project HEARTH GitHub Demo — Oct 2026 – Nov 2026
• Developed a NASA fire-safety research dashboard with an AI tool-using agent, evidence-grounded insights, interactive charts, and experiment analysis using Next.js, TypeScript, and Tailwind CSS.

WashCraft Property Maintenance GitHub Demo — Oct 2026 – Nov 2026
• Built a full-stack business website with lead management and an admin panel, integrated Neon PostgreSQL and Cloudflare R2, and deployed on Vercel.

Agentic Safety & Security Surveillance System GitHub Demo — Aug 2026 – Sep 2026
• Built a FastAPI and React video-analysis module using YOLO11s and ByteTrack for person detection and tracking, with annotated videos, event summaries, and email alerts.

Bangla Voice Assistant for Online Forms GitHub — Jun 2026 – Jul 2026
• Developed a Chrome extension with Bangla and English voice input, LLM-based translation, and automated form filling; added field validation, read-back verification, and user-confirmed submission.

MemeGuard AI GitHub Demo — Jan 2026 – Apr 2026
• Built a real-time multimodal hate-speech moderation platform using ResNet-18 and DistilBERT, with Flask, REST APIs, and WebSockets for web and Telegram integration.

RESEARCH PROJECTS

Distilling Hate: Lightweight Transformer Models for Fine-Grained Bengali Hate Speech Detection GitHub
Published in the PLOS ONE journal.
• Developed a lightweight Bengali hate speech detection model by distilling BanglaBERT Large (336.7M parameters) into a 13.7M-parameter student using cross-entropy, KL divergence, and hidden-state alignment.

JSSL-GI: Joint Self-Supervised Learning for Multi-Category Gastrointestinal Endoscopy Image Classification GitHub
Manuscript Under Review at IEEE Access.
• Developed a joint self-supervised learning pipeline using an ImageNet-pretrained ViT-Small, combining masked autoencoder reconstruction and multi-view contrastive learning with adaptive uncertainty-based loss weighting.

EDUCATION

B.Sc. in Computer Science and Engineering — Spring 2022 – Spring 2026
North South University

Higher Secondary Certificate — 2020 – 2021
Chittagong University College

Secondary School Certificate — 2017 – 2018
Hathazari Parboti Govt. High School

ACHIEVEMENTS & LEADERSHIP

Lead Coordinator — IEEE NSU IAS Student Branch Chapter — 2023 – 2025
• Led 24+ technical and professional development programs, coordinating logistics, budgets, speakers, sponsors, and faculty as part of the Events & Management Team.
• Received the Best Lead Award (2024–2025) for leadership and team coordination.